

1. $A = 3x^2 + x - 1$, $B = 3x^2 + x + 2$, $C = 2x^2 + 5x + 3$ であるとき，次の計算をせよ。

(1) $A + C$

答. _____

(2) $2A - 3C$

答. _____

(3) $A - B - C$

答. _____

(4) $A - B + C$

答. _____

(5) $A - 3B - 2C$

答. _____

(6) $-A + 2B + C$

答. _____

(7) $A + B - 3C$

答. _____

(8) $3A - 2B - C$

答. _____

2. 次の 1 次方程式を解け。

(1) $-7x + 8 = -4x + 2$

答. _____

(2) $5x + 5 = 8x + 6$

答. _____

(3) $8x - 2(6x - 1) = -7$

答. _____

(4) $-3x - 8 + 2(-4x + 3) = 0$

答. _____

(5) $8x + 5(3x + 1) = -7$

答. _____

(6) $\frac{5}{3}x - 2 = -2x + 1$

答. _____

(7) $\frac{7}{4}x - \frac{1}{4} = 5x + \frac{8}{3}$

答. _____

(8) $\frac{1}{4}x - 3 = \frac{4}{3}x - 1$

答. _____

1. $A = 3x^2 + x - 1$, $B = 3x^2 + x + 2$, $C = 2x^2 + 5x + 3$ であるとき, 次の計算をせよ。

(1) $A + C$

答. $5x^2 + 6x + 2$

(2) $2A - 3C$

答. $-13x - 11$

(3) $A - B - C$

答. $-2x^2 - 5x - 6$

(4) $A - B + C$

答. $2x^2 + 5x$

(5) $A - 3B - 2C$

答. $-10x^2 - 12x - 13$

(6) $-A + 2B + C$

答. $5x^2 + 6x + 8$

(7) $A + B - 3C$

答. $-13x - 8$

(8) $3A - 2B - C$

答. $x^2 - 4x - 10$

2. 次の 1 次方程式を解け。

(1) $-7x + 8 = -4x + 2$

$-3x = -6$

答. $x = 2$

(2) $5x + 5 = 8x + 6$

$-3x = 1$

答. $x = -\frac{1}{3}$

(3) $8x - 2(6x - 1) = -7$

$8x - 12x + 2 = -7$

$-4x = -9$

答. $x = \frac{9}{4}$

(4) $-3x - 8 + 2(-4x + 3) = 0$

$-3x - 8 - 8x + 6 = 0$

$-11x = 2$

答. $x = -\frac{2}{11}$

(5) $8x + 5(3x + 1) = -7$

$8x + 15x + 5 = -7$

$23x = -12$

答. $x = -\frac{12}{23}$

(6) $\frac{5}{3}x - 2 = -2x + 1$

両辺に 3 をかけて, $5x - 6 = -6x + 3$

答. $x = \frac{9}{11}$

(7) $\frac{7}{4}x - \frac{1}{4} = 5x + \frac{8}{3}$

両辺に 12 をかけて, $21x - 3 = 60x + 32$

答. $x = -\frac{35}{39}$

(8) $\frac{1}{4}x - 3 = \frac{4}{3}x - 1$

両辺に 12 をかけて, $3x - 36 = 16x - 12$

答. $x = -\frac{24}{13}$